



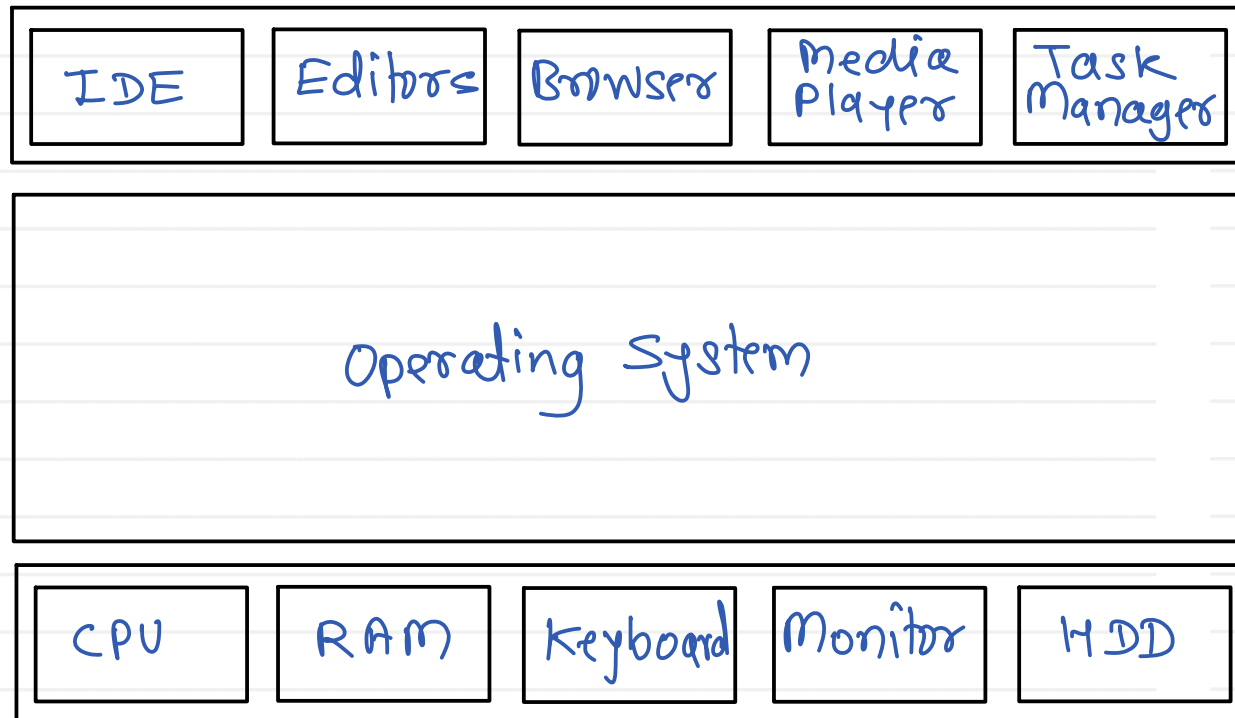
Sunbeam Institute of Information Technology
Pune and Karad

Operating System Concepts and Linux Programming

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End User



- interface between end user and hardware

Application S/W

- interface between Application S/W and computer H/W

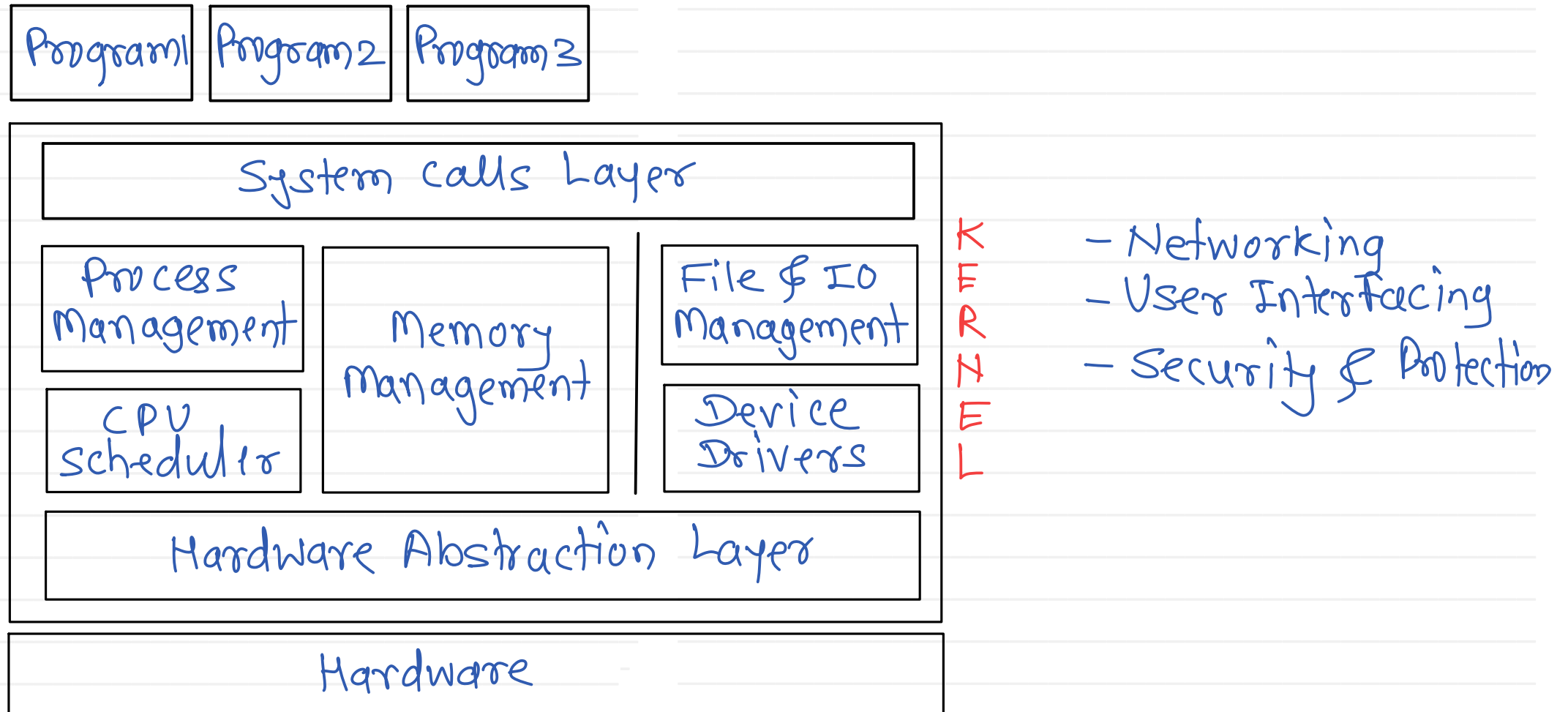
- control program which controls the execution of all other programs running inside a system.

- resource allocator/manager which manages all limited hardware resources

Computer H/W

CD/DVD/ISO = Core OS program (kernel) + System Utilities + Application S/W

Kernel functions



File *fp = fopen();



int fd = open();



sys_open();



Hardware (HDD)

- Library function

- System call API

} user
library

User
mode

- system call

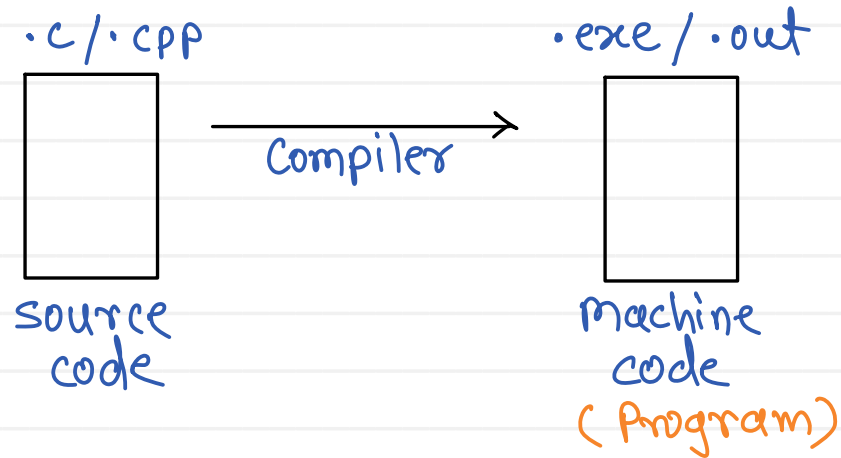
kernel
mode

Program compilation steps

GCC - GNU C Compiler
GNU - GNU NOT UNIX

Process : Program in execution

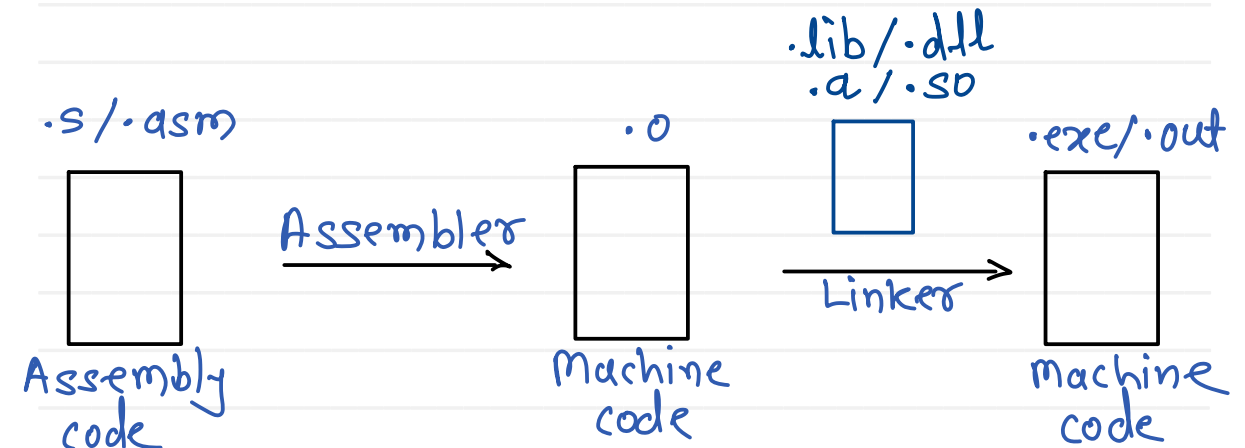
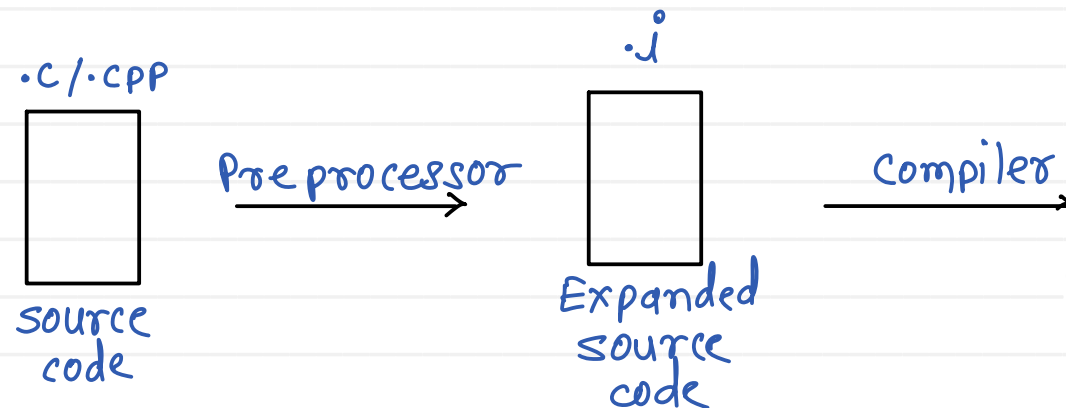
Program : set of instructions to machine (CPU)

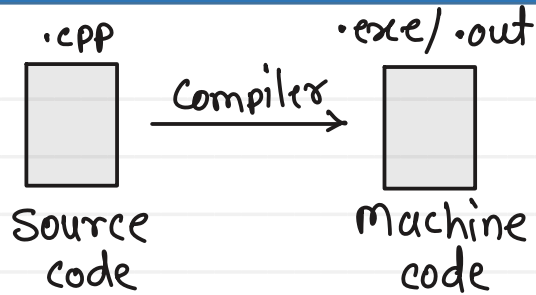


Toolchain : (gcc)

- set of tools which work on source code one by one to convert it into machine code.

1. Preprocessor (cpp)
2. Compiler (cc)
3. Assembler (as)
4. Linker (ld)
5. Debugger (gdb)
6. Utilities (make, objdump)





- information required to execute a program
- magic number (2 or 4 bytes)
 - identity to file format

Windows : .exe → Portable Executable

↓
MZ

Linux : .out → Executable Linking Format → ELF (sectioned binary)

- type of a program → (GUI/CLI/library)
- info about remaining sections → (start, end, size)
- address of entry point function

Program
.exe/.out

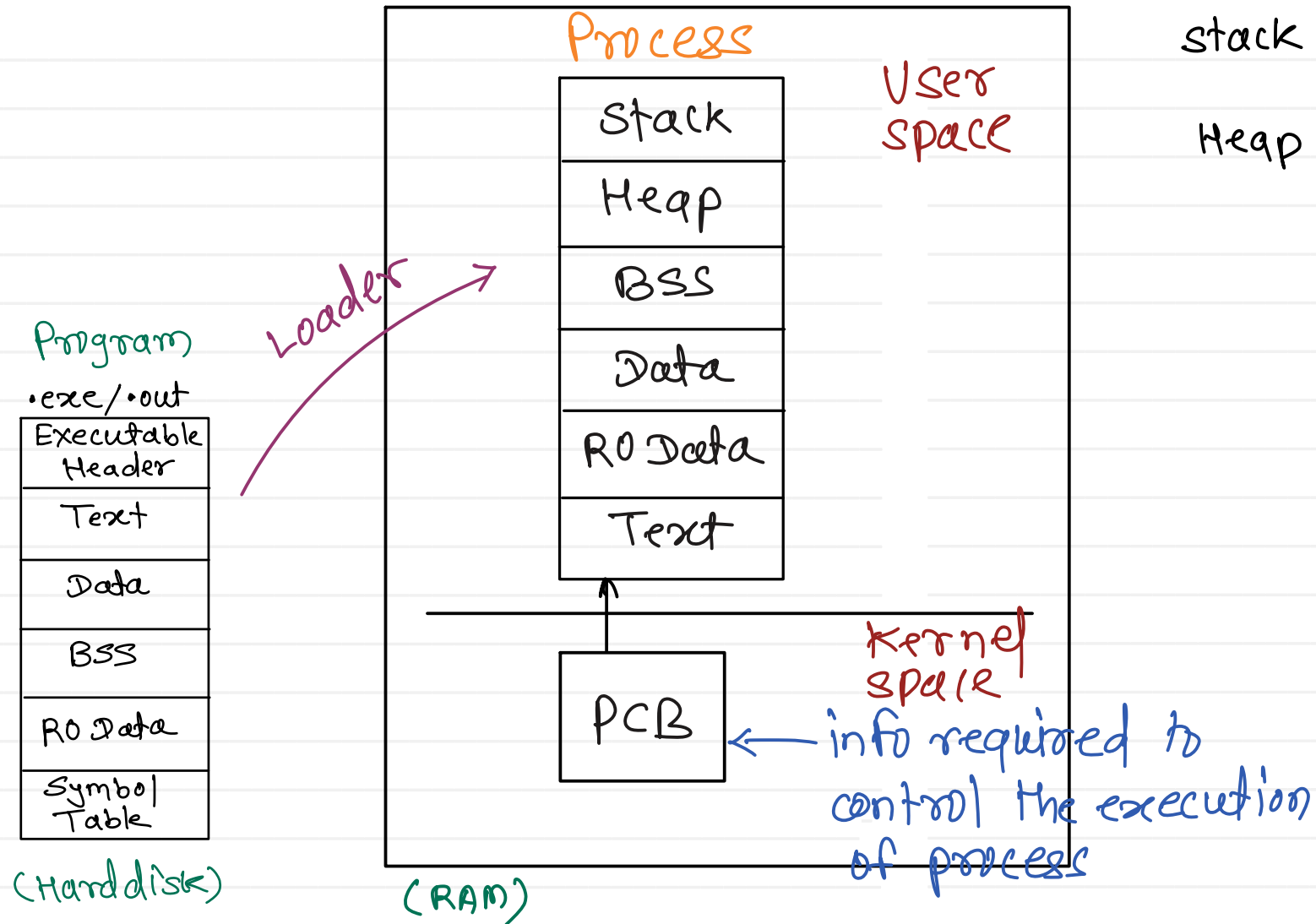
Exe Header
.text
.rodata
.data
.bss
.symtab
.symstr

- instruction in machinecode format
- read only data (string constant)
- initialized static & global variables
- un initialized static & global variables
- Symbol table

- information about symbols of a program
 symbols <

- variables (name, addr, section, size, default value)
- functions (name, addr, return type, no./type args)

 → strings which are created by compiler for all the symbols.



stack - FARs are stored
 ↳ return addr, local variables
 Heap - dynamically allocated space

PCB - process descriptor
 ↓
 struct task_struct
 (sched.h)

- pid, ppid
- exit status
- Mem info
- Sched info
- IPC info
- Open file info
- execution context
- Kernel stack

File : collection of a data

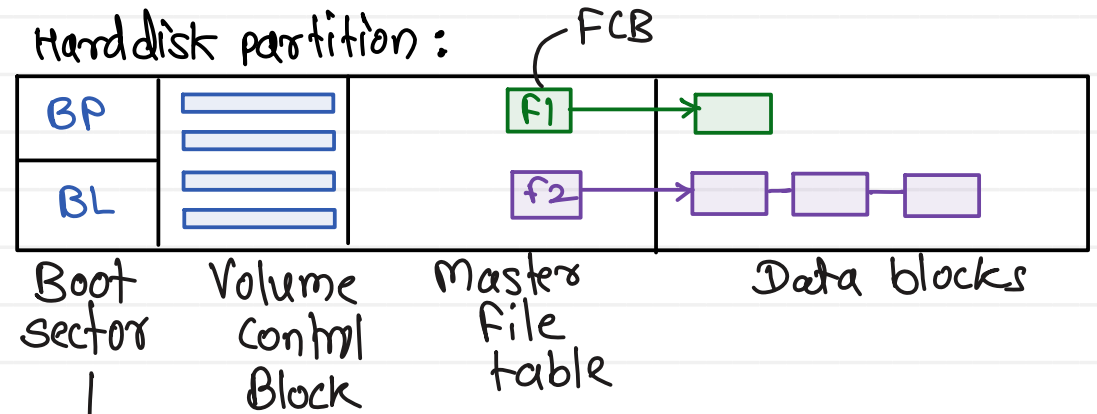
File : data + metadata
(actual content of a file) (information about a file)

↓
data block

↓
File control Block (FCB)

- name
- user/owner, group
- size
- permissions
r=read, w=write, x=execute
user/group/others
- timestamp
created/accessed/modified
- no. of links
- type
- info about data blocks

File system : Organizing files on a partition



- ↓
- contains two programs which are used at the time of booting
 1. Bootstrap program
 2. Boot loader

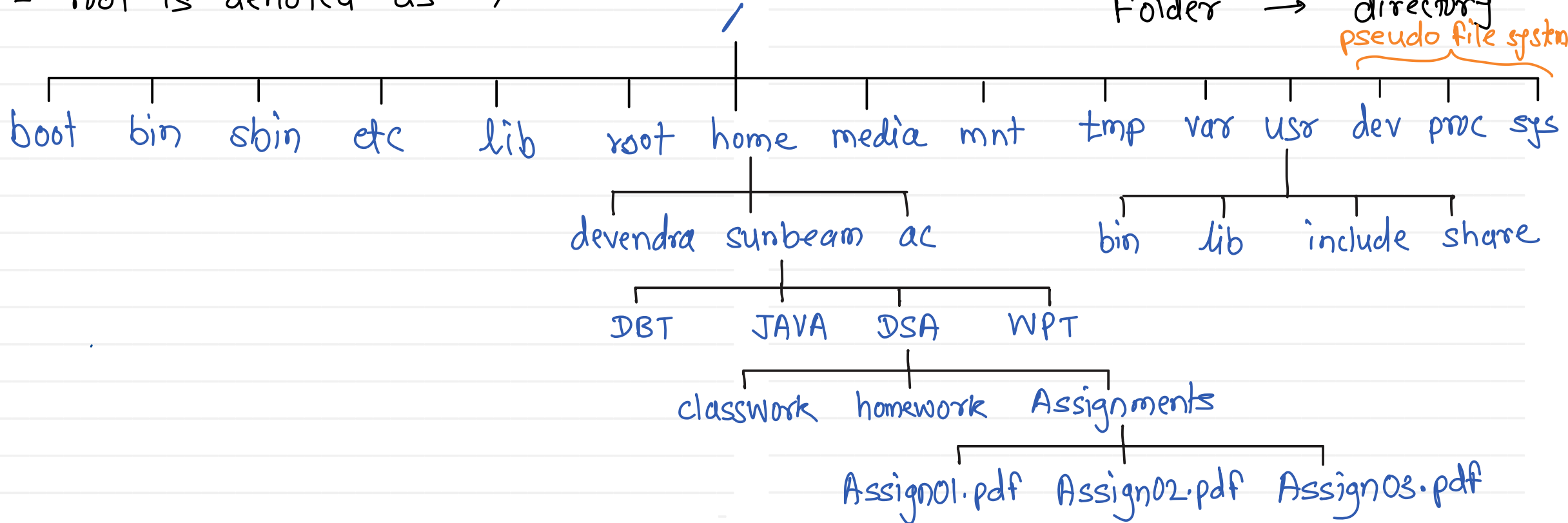
- ↓
- info about partition
 - label
 - size
 - empty, filled
 - info about free data blocks of a partitions

Linux file structure

- Linux follows root file structure, which starts at root.
- root is denoted as '/'

Administrator → super user
admin → root
file → file
Folder → directory

pseudo file systems



	Windows	Linux
static	.lib	.a ← archive
dynamic	.dll	.so ← shared object

libc.so

gcc main.c ← dynamic linking
gcc -static main.c ← static linking

1. Absolute path :

- always starts with /

/home/sunbeam/DSA/Assignments/Assign02.pdf

2. Relative path :

- always starts with respect to working dir
present/current working dir = /home/sunbeam
DSA/Assignments/Assign02.pdf

